

Algebra: Solving Linear Systems by Graphing

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Name: _____ Date: _____ Score: _____

DIRECTIONS Identify the solution (intersection point) of each system. Write as an ordered pair, or state "No Solution" or "Infinitely Many."**1** Solve by graphing:

$$y = x + 1$$
$$y = -x + 3$$

Answer: _____

2 Solve by graphing:

$$y = 2x - 1$$
$$y = x + 1$$

Answer: _____

3 Solve by graphing:

$$y = 3x$$
$$y = x + 4$$

Answer: _____

4 Solve by graphing:

$$y = x - 2$$
$$y = -2x + 4$$

Answer: _____

5 Solve by graphing:

$$y = 2x + 1$$
$$y = 2x - 3$$

Answer: _____

6 Solve by graphing:

$$y = x + 2$$
$$y = -x + 6$$

Answer: _____

7 Solve by graphing:

$$y = -x + 5$$
$$y = 2x - 1$$

Answer: _____

8 Solve by graphing:

$$y = (1/2)x + 1$$
$$y = x - 1$$

Answer: _____

9 Solve by graphing:

$$y = 3x - 2$$
$$y = 3x - 2$$

Answer: _____

10 Solve by graphing:

$$y = -2x + 4$$
$$y = x - 2$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES

Graph both lines using slope-intercept form. Intersection = solution. Parallel lines = no solution. Same line = infinite

ly many.

1 $y=x+1$ and $y=-x+3$
= (1, 2)

Set equal: $x+1=-x+3$, $2x=2$, $x=1$, $y=2$

2 $y=2x-1$ and $y=x+1$
= (2, 3)

$2x-1=x+1$, $x=2$, $y=3$

3 $y=3x$ and $y=x+4$
= (2, 6)

$3x=x+4$, $2x=4$, $x=2$, $y=6$

4 $y=x-2$ and $y=-2x+4$
= (2, 0)

$x-2=-2x+4$, $3x=6$, $x=2$, $y=0$

5 $y=2x+1$ and $y=2x-3$
= No Solution

Same slope, different y-int $\rightarrow$ parallel lines

6 $y=x+2$ and $y=-x+6$
= (2, 4)

$x+2=-x+6$, $2x=4$, $x=2$, $y=4$

7 $y=-x+5$ and $y=2x-1$
= (2, 3)

$-x+5=2x-1$, $6=3x$, $x=2$, $y=3$

8 $y=1/2x+1$ and $y=x-1$
= (4, 3)

$1/2x+1=x-1$, $2=1/2x$, $x=4$, $y=3$

9 $y=3x-2$ and $y=3x-2$
= Infinitely Many

Identical equations $\rightarrow$ same line

10 $y=-2x+4$ and $y=x-2$
= (2, 0)

$-2x+4=x-2$, $6=3x$, $x=2$, $y=0$

